

MUSSALA, Bulgaria

WMO: 156150

Lat: 42.1793N

Lon: 23.5851E

Elev: 2927

StdP: 70.76

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -21.9	(c) -19.8	(d) -29.8	(e) 0.3	(f) -13.5	(g) -26.3	(h) 0.5	(i) -15.1	(j) 34.1	(k) -7.1	(l) 28.4	(m) -7.5	(n) 9.5	(o) 20	(p) 1.025

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.0	(c) 13.9	(d) 8.2	(e) 12.4	(f) 7.5	(g) 11.1	(h) 6.9	(i) 9.4	(j) 12.1	(k) 8.5	(l) 11.1	(m) 7.7	(n) 10.1	(o) 4.1	(p) 230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 8.2	(b) 9.7	(c) 10.3	(d) 7.3	(e) 9.1	(f) 9.4	(g) 6.5	(h) 8.6	(i) 8.6	(j) 36.0	(k) 12.3	(l) 33.4	(m) 11.2	(n) 31.3	(o) 10.2	(p) 15.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	20.1	16.1	-24.4	16.6	2.8	1.6	-26.4	17.7	-28.1	18.7	-29.7	19.6	-31.7	20.7
			-24.7	11.8	2.8	1.5	-26.7	12.9	-28.4	13.8	-29.9	14.7	-31.9	15.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-9.7	-9.5	-7.9	-4.5	0.0	3.8	6.3	6.7	3.0	-0.1	-4.2	-7.9		
	DBStd	7.06	4.65	4.70	4.17	3.79	3.21	3.57	3.19	2.96	3.53	4.14	4.90	4.55		
	HDD10.0	4376	609	547	554	434	311	187	122	108	210	312	427	556		
	HDD18.3	7403	868	781	813	684	569	435	374	360	459	570	677	814		
	CDD10.0	14	0	0	0	0	0	2	6	7	1	0	0	0		
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	7.0	8.8	9.0	8.9	7.6	6.3	5.4	4.8	4.4	5.3	6.7	8.6	9.0		
(15) Precipitation	PrecAvg	794	69	71	84	92	88	71	67	52	52	44	41	65		
	PrecMax	1011	170	208	190	253	149	155	140	133	166	104	124	137		
	PrecMin	541	21	12	30	40	35	12	14	8	10	6	0	5		
	PrecStd	142	40	47	39	45	30	31	33	36	31	25	26	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.2	1.6	2.3	5.7	8.8	14.4	16.4	15.6	12.6	10.7	6.6	3.7		
		MCWB	-3.8	-2.8	-1.2	1.1	3.8	7.0	8.9	9.4	6.1	4.5	0.6	-2.1		
	2%	DB	-0.8	-0.9	0.1	3.7	7.2	12.1	14.1	13.8	10.6	8.4	4.8	0.4		
		MCWB	-4.8	-3.9	-2.4	0.3	3.1	6.8	8.4	8.6	5.5	2.8	-0.6	-3.4		
	5%	DB	-2.4	-2.5	-1.4	1.7	5.7	10.4	12.4	12.4	9.2	6.6	3.1	-1.0		
		MCWB	-5.3	-4.8	-3.3	-0.7	2.7	6.2	7.8	7.8	4.8	2.1	-1.4	-3.7		
	10%	DB	-3.9	-3.9	-2.6	0.1	4.3	8.9	11.0	11.0	7.7	5.1	1.5	-2.3		
		MCWB	-5.9	-5.8	-4.1	-1.2	2.0	5.6	7.1	7.1	4.2	1.1	-2.2	-4.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.1	-1.9	-0.3	2.3	5.3	9.1	11.2	11.1	7.8	6.1	2.6	-0.6		
		MCDB	-0.4	-0.4	0.8	4.5	7.5	11.6	13.8	13.7	10.4	8.9	4.8	0.5		
	2%	WB	-3.8	-3.2	-1.7	0.6	4.0	8.0	9.5	9.5	6.6	4.5	0.6	-1.7		
		MCDB	-2.4	-1.7	-0.6	2.7	6.0	10.5	12.1	12.4	9.3	6.9	2.5	-0.5		
	5%	WB	-4.8	-4.3	-2.8	-0.3	3.2	7.1	8.5	8.6	5.8	3.3	-0.4	-2.9		
		MCDB	-3.2	-2.8	-1.8	1.1	5.0	9.2	11.1	11.3	8.0	5.1	1.2	-1.7		
	10%	WB	-5.7	-5.4	-3.9	-1.0	2.4	6.2	7.7	7.7	5.0	2.2	-1.2	-4.1		
		MCDB	-4.2	-4.3	-2.9	0.1	3.9	8.3	10.1	10.2	7.0	3.7	0.6	-3.0		
(35) Mean Daily Temperature Range	MDBR		4.6	4.7	4.8	4.6	4.1	4.6	5.1	5.0	4.5	4.4	4.4	4.3		
	5% DB	MCDBR	5.3	5.1	5.1	5.3	5.4	6.3	6.9	6.5	6.0	5.9	5.0	4.6		
		MCWBR	4.5	4.3	4.5	4.1	4.5	5.5	5.5	5.6	5.1	5.1	3.7	3.6		
	5% WB	MCDBR	5.0	4.9	4.8	4.6	4.9	5.8	6.1	6.0	5.4	5.3	4.1	4.1		
		MCWBR	4.7	4.6	4.6	4.1	4.4	5.6	5.5	5.6	4.8	5.1	3.8	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.222	0.239	0.268	0.305	0.304	0.302	0.302	0.307	0.291	0.265	0.242	0.225		
	taud		2.525	2.513	2.500	2.421	2.459	2.476	2.466	2.447	2.494	2.580	2.617	2.560		
	Ebn at Noon		966	992	989	965	969	967	964	951	949	946	933	934		
	Edh at Noon		67	81	94	110	109	108	108	106	94	76	62	60		
(44) All-Sky Solar Radiation	RadAvg		1.68	2.50	3.52	4.52	5.41	6.10	6.36	5.76	4.27	2.97	1.94	1.44		
	RadStd		0.25	0.32	0.37	0.52	0.40	0.52	0.46	0.47	0.34	0.35	0.19	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.61	N/A	N/A	N/A	+0.56	N/A	-227	-234	N/A	N/A
Regional (30 neighbors)		+0.47	N/A	N/A	N/A	+0.40	+0.40	N/A	-142	N/A	N/A

Nomenclature: See separate page